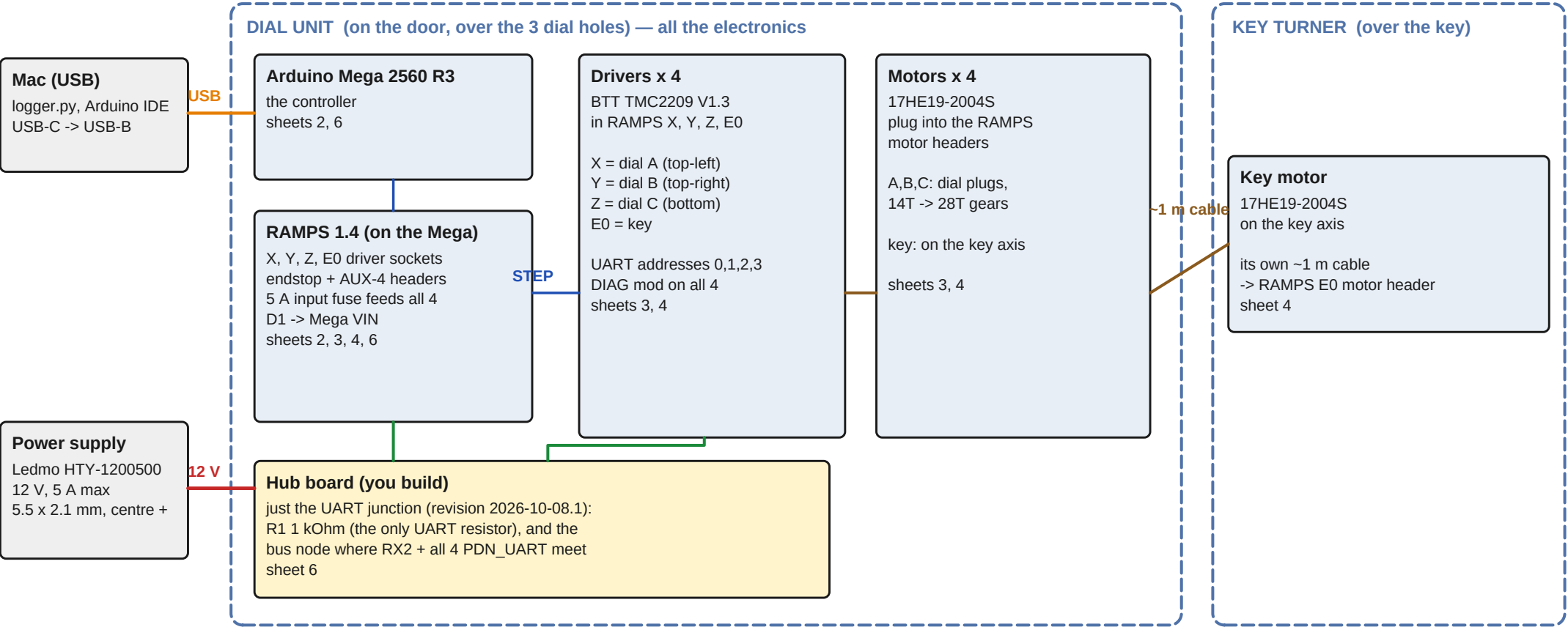


Overview

What connects to what. Each block is drawn in full on the sheet named in it.



Sheets

- 1 Overview (this sheet)
- 2 Power: 12 V, 5 V, GND
- 3 Drivers A and B on the RAMPS
- 4 Driver C and the key driver (E0)
- 5 DIAG-mod and UART-tap details
- 6 Mega + RAMPS headers, hub (UART bus)
- 7 Pin map, jumpers, currents, parts

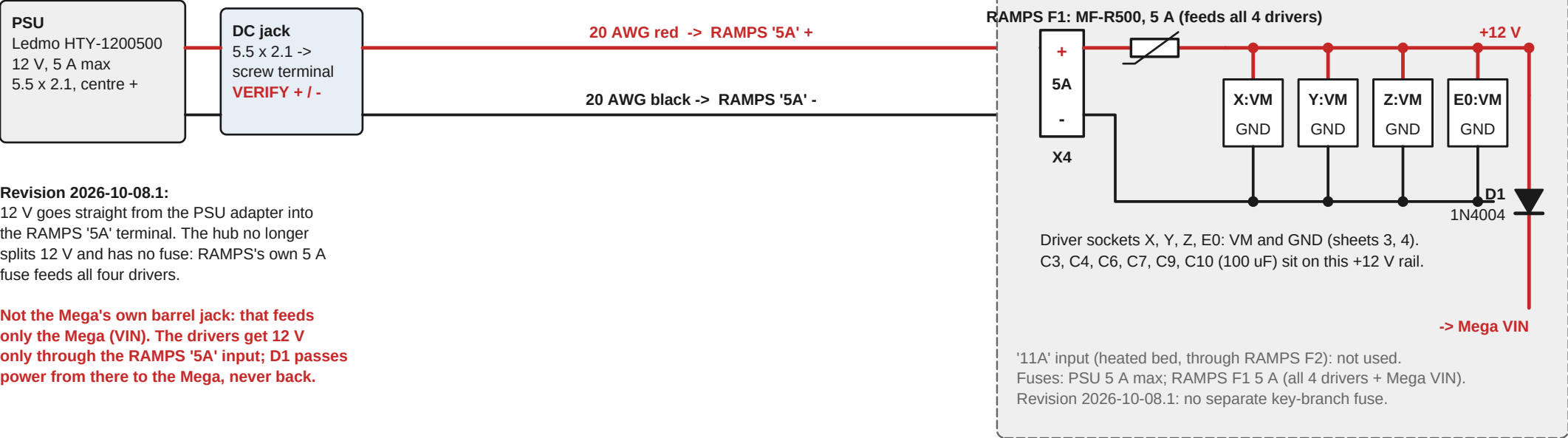
Rules

Make every connection with power off (USB unplugged, 12 V off).
Never plug or unplug a motor with 12 V on (the key motor cable included).
Power up: USB first, then 12 V. Power down: 12 V first.
Never fit an MS3 jumper: on the V1.3 that pin is the UART line.
Emergency stop: pull the 12 V plug. `!` aborts a move.
VERIFY = not yet checked on the bench (control/bringup.md).
Revision 2026-10-08.1: key driver in E0, no remote board or cable.
If this disagrees with control/wiring.md, wiring.md wins.

- +12 V — +5 V (logic, VIO) — GND — UART bus
- STEP / DIR / EN, button — DIAG (stall) — motor coils — jumper / tie / spare
- bought module — board you build — already on the RAMPS / Mega: nothing to buy

Power: 12 V, 5 V, GND

12 V goes from the PSU straight into the RAMPS '5A' input (its own 5 A fuse feeds all four drivers). 5 V logic comes from the Mega.

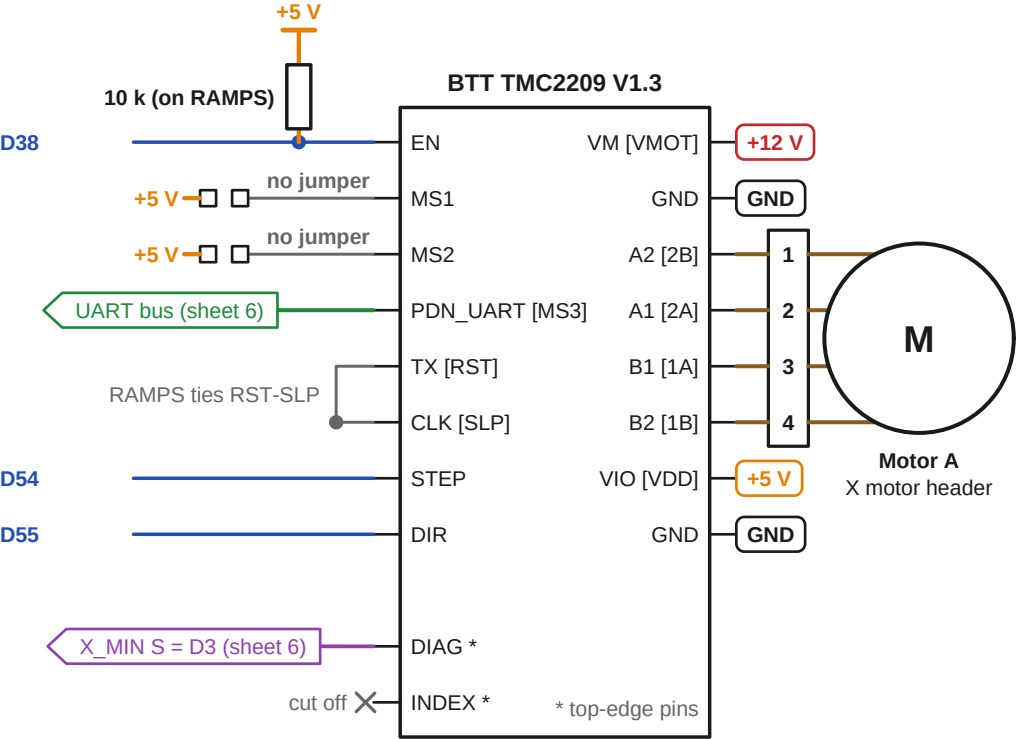


Drivers A and B on the RAMPS

[] = the RAMPS socket name of that pin. All connections to the left are on the RAMPS or by jumper.

Socket X - dial A (top-left)

UART address 0 - BTT TMC2209 V1.3 with the DIAG mod

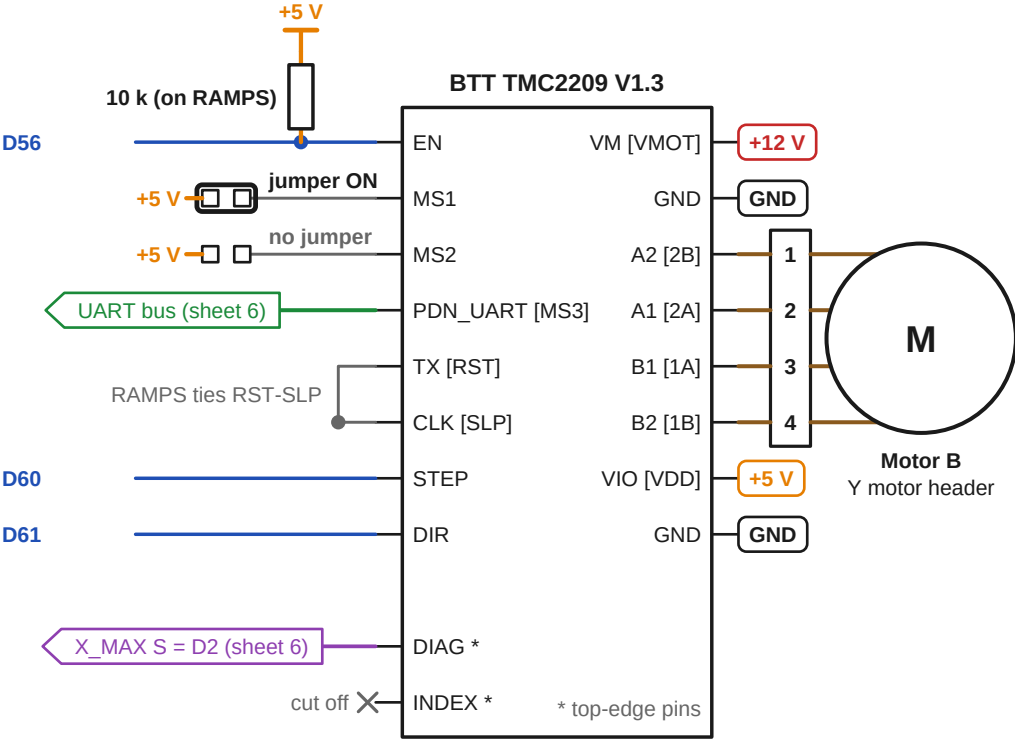


Jumpers under socket X: MS1 off, MS2 off, MS3 off (never fit MS3)

UART lead: clipped onto the MS3 jumper pin, signal side (sheet 5).
STEP D54 - DIR D55 - EN D38. The RAMPS pull-up keeps the driver off until the firmware pulls EN low.
DIAG lead (the mod, sheet 5) -> X_MIN S pin: D3, an interrupt pin.
Motor 17HE19-2004S: header pins 1-2 = one coil, 3-4 = the other.

Socket Y - dial B (top-right)

UART address 1 - BTT TMC2209 V1.3 with the DIAG mod



Jumpers under socket Y: MS1 ON, MS2 off, MS3 off (never fit MS3)

UART lead: clipped onto the MS3 jumper pin, signal side (sheet 5).
STEP D60 - DIR D61 - EN D56. The RAMPS pull-up keeps the driver off until the firmware pulls EN low.
DIAG lead (the mod, sheet 5) -> X_MAX S pin: D2, an interrupt pin.
Motor 17HE19-2004S: header pins 1-2 = one coil, 3-4 = the other.

[] = the RAMPS / StepStick socket name of that pin. Fit each driver with EN, DIR, VM and GND matching the RAMPS silkscreen: a reversed driver is destroyed.
The BTT V1.3 puts PDN_UART in the socket's MS3 position. TX: R10 is not fitted, so TX connects to nothing. Don't bridge R10.
CLK: the RAMPS ties the socket's RST and SLP pins (= TX and CLK on the V1.3). With R10 open, CLK sees only its 20 k pull-down -> internal clock.
Motor currents are set over UART: 1.0 A RMS, hold 0.5 A. If a motor buzzes instead of turning, swap the middle two wires in its plug.

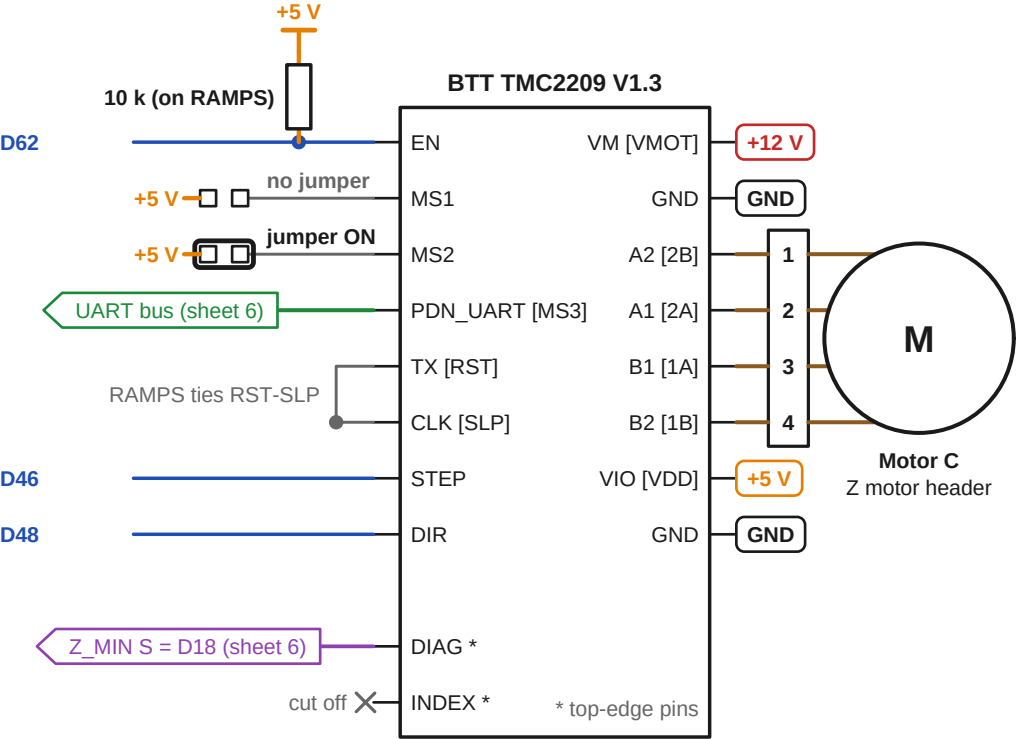
- +12 V
- +5 V (logic, VIO)
- GND
- UART bus
- STEP / DIR / EN, button
- DIAG (stall)
- motor coils
- jumper / tie / spare
- bought module
- board you build
- already on the RAMPS / Mega: nothing to buy

Dial driver C and the key driver (E0)

The key driver is an ordinary driver in the E0 socket (revision 2026-10-08.1).

Socket Z - dial C (bottom)

UART address 2 - BTT TMC2209 V1.3 with the DIAG mod



Jumpers under socket Z: MS1 off, MS2 ON, MS3 off (never fit MS3)

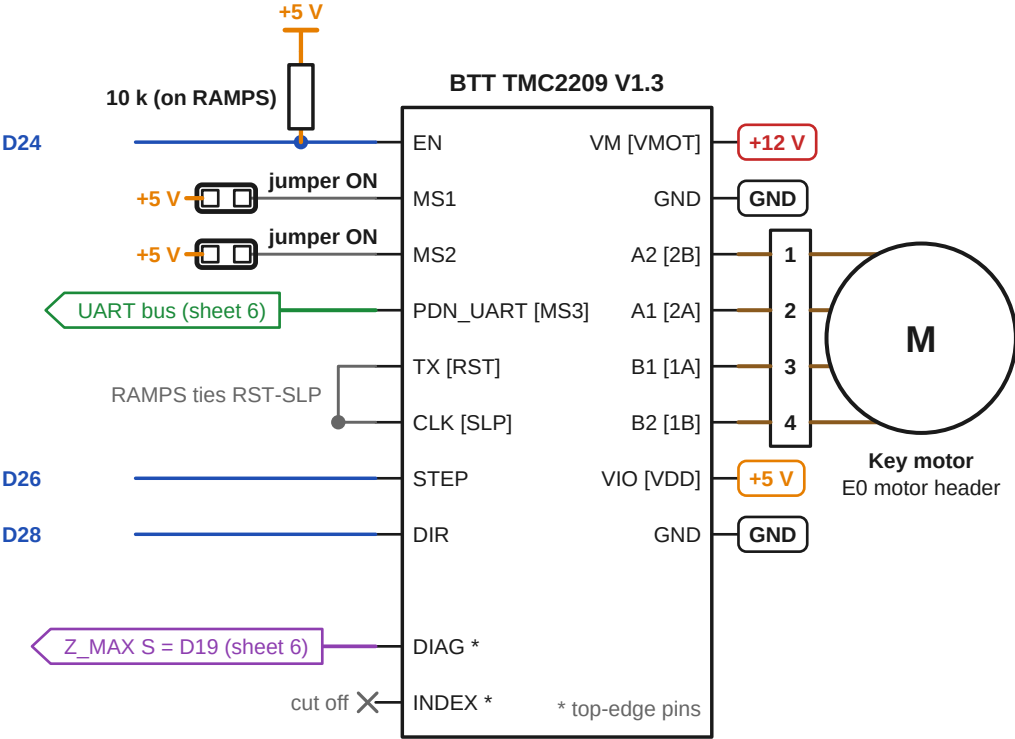
UART lead: clipped onto the MS3 jumper pin, signal side (sheet 5).
STEP D46 - DIR D48 - EN D62. The RAMPS pull-up keeps the driver off until the firmware pulls EN low.
DIAG lead (the mod, sheet 5) -> Z_MIN S pin: D18, an interrupt pin.
Motor 17HE19-2004S: header pins 1-2 = one coil, 3-4 = the other.

Revision 2026-10-08.1: the key driver is an ordinary driver in the RAMPS E0 socket, like the dials.

E0 address 3 = MS1 + MS2 jumpers ON. The key's STEP/DIR/EN are D26/D28/D24 (Marlin pins_RAMPS.h E0_STEP/DIR/ENABLE).
The key motor's own ~1 m cable plugs into the E0 motor header. Use it as supplied; shorten it only if the key's StallGuard reads too dull (bench stage 5).
Key current: 0.6 A RMS to start, hold 0.3 A. E0's MS3 jumper pin and DIAG mod are VERIFY items (bench stages 1, 3) like the dials'.

Socket E0 - the key turner

UART address 3 - BTT TMC2209 V1.3 with the DIAG mod



Jumpers under socket E0: MS1 ON, MS2 ON, MS3 off (never fit MS3)

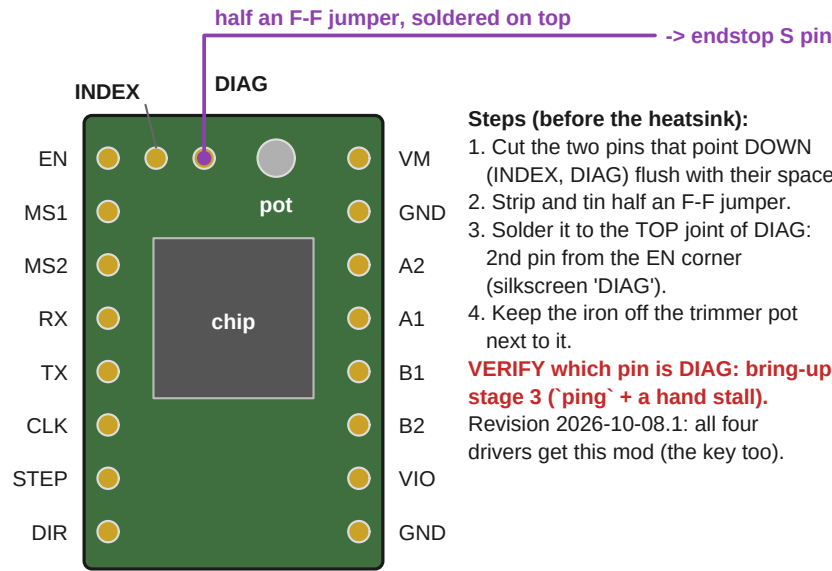
UART lead: clipped onto the MS3 jumper pin, signal side (sheet 5).
STEP D26 - DIR D28 - EN D24. The RAMPS pull-up keeps the driver off until the firmware pulls EN low.
DIAG lead (the mod, sheet 5) -> Z_MAX S pin: D19, an interrupt pin.
Motor 17HE19-2004S: header pins 1-2 = one coil, 3-4 = the other.

+12 V +5 V (logic, VIO) GND UART bus
STEP / DIR / EN, button DIAG (stall) motor coils jumper / tie / spare
bought module board you build already on the RAMPS / Mega: nothing to buy

DIAG-mod and UART-tap details; the key motor cable

DIAG mod (all four drivers)

BTT TMC2209 V1.3 seen from above - not to scale



Steps (before the heatsink):

1. Cut the two pins that point DOWN (INDEX, DIAG) flush with their spacer.
2. Strip and tin half an F-F jumper.
3. Solder it to the TOP joint of DIAG: 2nd pin from the EN corner (silkscreen 'DIAG').
4. Keep the iron off the trimmer pot next to it.

VERIFY which pin is DIAG: bring-up stage 3 (`ping` + a hand stall).
Revision 2026-10-08.1: all four drivers get this mod (the key too).

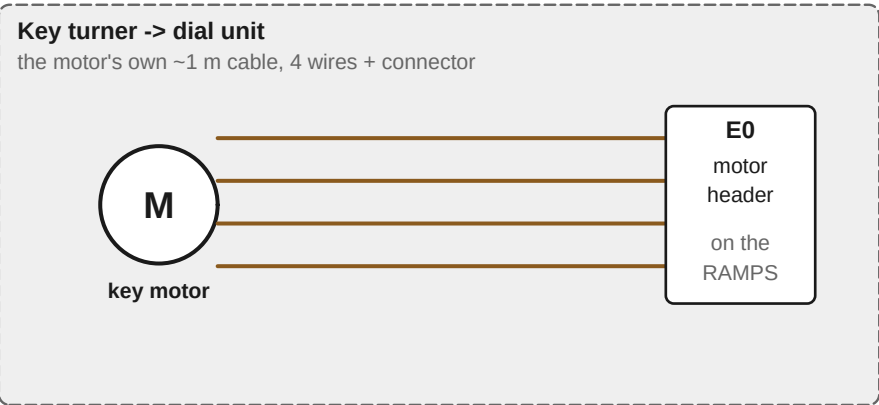
UART tap: the MS3 jumper pin (each socket)

RAMPS jumper block under a driver socket - not to scale



Key motor cable (the only wire between the units)

revision 2026-10-08.1



Plugs into the E0 motor header like a dial motor.
Use the supplied ~1 m cable as it is; shorten it only if the key's StallGuard reads too dull to find the stop (bench stage 5).
Never plug or unplug it with 12 V on.
If the motor buzzes instead of turning, swap its middle two wires.

Revision 2026-10-08.1 replaced the 6-conductor inter-unit signal cable and the remote driver board with this one motor cable.

+12 V +5 V (logic, VIO) GND UART bus

STEP / DIR / EN, button DIAG (stall) motor coils jumper / tie / spare

bought module board you build already on the RAMPS / Mega: nothing to buy

Thin lines = RAMPS traces. Thick lines = your jumper wires (F-F, 10-20 cm) and hub-board wiring.



Pin map, jumpers, currents, parts

Pin map (control/wiring.md 1; Marlin pins_RAMPS.h) - revision 2026-10-08.1

Function	Mega pin	Where it connects
Dial A STEP / DIR / EN	D54 / D55 / D38	RAMPS X socket
Dial B STEP / DIR / EN	D60 / D61 / D56	RAMPS Y socket
Dial C STEP / DIR / EN	D46 / D48 / D62	RAMPS Z socket
Key STEP / DIR / EN	D26 / D28 / D24	RAMPS E0 socket
Dial A DIAG	D3 (interrupt)	X_MIN header, S
Dial B DIAG	D2 (interrupt)	X_MAX header, S
Dial C DIAG	D18 (interrupt)	Z_MIN header, S
Key DIAG	D19 (interrupt)	Z_MAX header, S (E0 DIAG mod)
UART TX2	D16	AUX-4 pin 18 -> hub R1
UART RX2	D17	AUX-4 pin 17 -> hub bus
Drivers' PDN_UART	-	MS3 jumper pins X,Y,Z,E0 -> bus
Start / stop button	D14 (pull-up)	Y_MIN header, S and -
Status LED	D13	on-board LED 'L'
Spare	D15 / D23	Y_MAX S / AUX-4 16
USB serial (the log)	D0 / D1	USB: keep free

Power order (8)

1. Make every connection with power off (USB unplugged, 12 V off).
2. Before the first power-up: every driver's VREF pot to minimum.
3. Power up: USB first, then 12 V. 4. Power down: 12 V first, then USB.
5. Never plug or unplug a motor with 12 V on (the key motor cable too).
6. Emergency stop: pull the 12 V plug.

VERIFY on the bench (control/bringup.md)

Which MS3 jumper pin is the signal side (stage 1, `ping`), all four sockets.
Which top-edge pin is DIAG (stage 3). The key answers at address 3 (E0).
The key's StallGuard through the 1 m motor cable (stage 5); shorten if dull.
Multimeter: DC-jack + / - before the first power-up.
Sources: control/wiring.md - Marlin pins_RAMPS.h, TMC2209 datasheet rev 1.09,
BTT TMC2209 V1.3 manual + schematic, RAMPS 1.4 KiCad netlist.

+12 V

+5 V (logic, VIO)

GND

UART bus

STEP / DIR / EN, button

DIAG (stall)

motor coils

jumper / tie / spare

bought module

board you build

already on the RAMPS / Mega: nothing to buy

Driver jumpers and UART addresses (2)

Socket	Role	MS1	MS2	MS3	Address
X	A (top-left)	off	off	off	0
Y	B (top-right)	ON	off	off	1
Z	C (bottom)	off	ON	off	2
E0	key	ON	ON	off	3

Never fit an MS3 jumper. Socket E1 stays empty.

Motor currents, set over UART (7)

Driver	Run	Hold	Why
Dials A, B, C	1.0 A RMS	0.5 A	until the dial torque is measured
Key	0.6 A RMS	0.3 A	then 2 x the measured minimum, <= 1.0 A
Firmware ceiling	1.2 A	-	BTT: active cooling above 1.2 A

Parts you fit (5, 10) - full list: docs/bom.md

Where	Ref	Part	Status
Hub	R1	1 kOhm: the only UART resistor (TX2 -> bus)	have
Hub	-	male pins x 6: 5V BUS TX2 BUS BUS BUS (UART only)	have
PSU	-	DC jack 5.5 x 2.1 mm -> screw terminal, -> RAMPS 5A	ordered
PSU	-	20 AWG red + black, adapter -> RAMPS 5A terminal	ordered
RAMPS	S1	7 mm push button (start / stop)	have
Wiring	-	F-F jumpers, ~20 x 10-20 cm	ordered
Key	-	the motor's own ~1 m cable -> E0 motor header	have
Hub	-	perfboard (kit): a few cm square	have

Revision 2026-10-08.1 removed the remote driver board, the 6-conductor cable + Phoenix connectors, the hub 12 V branch + fuse, and C1a/C1b.
RAMPS F1, F2, D1, the six 100 uF and the 10 k pull-ups are on the RAMPS.